No-Arbitrage Principles and the Fundamental Theorems of Asset Pricing

Dominance, the law of one price, and the architecture behind martingale pricing

Lecture 8

One question drives the whole lecture

The question

Lecture 7 priced claims by conditional expectation under $\mathbb{Q}$. Where does that pricing measure come from, and how far can no-arbitrage take us before a model is chosen?

1. Revisit no-arbitrage precisely in a one-period market.
2. Derive price inequalities from the simplest possible argument: dominance.
3. Turn the law of one price into a replication principle.
4. Package no-arbitrage as existence of a consistent set of state prices — an equivalent martingale measure.
5. State the two Fundamental Theorems and watch them succeed, and fail, on small numerical examples.
6. Use dominance systematically to bound prices when exact replication is impossible.

A Formal Language for No-Arbitrage

Before any theorem, we need a precise market

Fix a finite set of states $\Omega = \{\omega_1, \ldots, \omega_K\}$, all with positive probability under $\mathbb{P}$, and traded assets $S^0, S^1, \ldots, S^m$ with today's prices $S_0^0, \ldots, S_0^m$ and time-T payoffs $S_T^i(\omega)$.

Trading strategy

A strategy is a vector $\theta = (\theta^0, \ldots, \theta^m) \in \mathbb{R}^{m+1}$, with value

$$V_0(\theta) = \sum_{i=0}^{m} \theta^i S_0^i, \qquad V_T(\theta, \omega) = \sum_{i=0}^{m} \theta^i S_T^i(\omega).$$

What exactly should “something for nothing” mean?

YOUR TURN State two ways a strategy can give a free lunch: one starting at zero cost, the other starting with money received today.

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Answer: two equivalent forms of arbitrage

- $V_0(\theta) = 0$, $V_T(\theta) \geq 0$ in every state and $V_T(\theta) > 0$ in at least one state;
- $V_0(\theta) < 0$ and $V_T(\theta) \geq 0$ in every state.

The bond converts the second form into a zero-cost strategy.

Can you turn a type-(ii) arbitrage into a type-(i) arbitrage?

YOUR TURN If $V_0(\theta) < 0$ and $V_T(\theta) \geq 0$, how can the initial receipt finance a bond position that brings today's cost to zero?

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Answer

Invest $-V_0(\theta) > 0$ in the bond. The combined cost is 0, and its terminal payoff is $V_T(\theta)$ plus a strictly positive bond payoff in every state.

The Dominance Principle

A payoff that is always at least as large should cost at least as much

Two claims mature at T, with payoffs $A \geq B$ in every state and prices a, b. Suppose $a < b$.

YOUR TURN Which claim do you buy and which do you sell? Track cash today and at T.

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Answer

Buy A, sell B: receive $b - a > 0$ today and obtain $A - B \geq 0$ at maturity. Thus no-arbitrage requires $\boxed{a \geq b}$.

Dominance is the most elementary monotonicity statement

$$A \geq B \text{ a.s.} \quad \Longrightarrow \quad \boxed{a \geq b.}$$

Reading

Larger payoff, larger price. No model, no distribution, no measure: this inequality holds contract by contract, state by state.

Where it is used

Every static price bound in this course — call and put bounds, superhedging, put–call inequalities — is dominance applied to a cleverly chosen pair of payoffs.

Which strike gives the more valuable call?

Two calls on the same stock, same maturity T, strikes $K_1 < K_2$.

YOUR TURN Without writing any formula, compare $(S_T - K_1)^+$ and $(S_T - K_2)^+$ state by state. Which call is worth more today, and why?

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$$(S_T - K_1)^+ \geq (S_T - K_2)^+ \quad \text{in every state,}$$

so dominance gives

$$\boxed{C_t(K_1) \geq C_t(K_2) \quad \text{whenever } K_1 < K_2.}$$

No model was needed: a lower strike can never be the cheaper call.

Does default risk make a bond cheaper, and can we prove it?

A government bond pays \$1 at T for certain. A corporate bond pays \$1 at T if the issuer does not default, and only a recovery amount $R < 1$ if it does.

YOUR TURN Compare the two payoffs state by state. What does dominance say about their prices, with no model of default probability at all?

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The corporate payoff is ≤ 1 in every state, matching the government bond exactly in the no-default state. Hence

$$\boxed{\text{price(corporate bond)} \leq \text{price(government bond)}.}$$

Must call prices be convex in the strike?

For strikes $K - h < K < K + h$, buy the outer calls and sell two calls at K.

YOUR TURN Find the butterfly payoff $f(S_T)$. Can it ever be negative?

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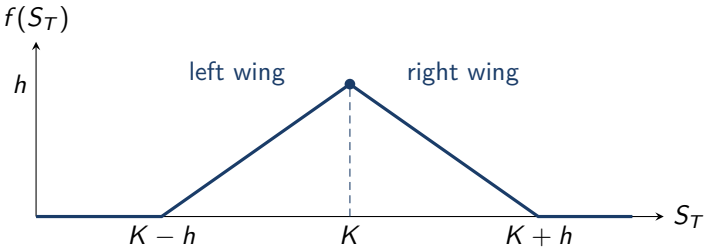
Answer

$$f(S_T) = (S_T - K + h)^+ - 2(S_T - K)^+ + (S_T - K - h)^+ = \big(h - |S_T - K|\big)^+ \geq 0.$$

Hence $\boxed{C_t(K-h) - 2C_t(K) + C_t(K+h) \geq 0}$.

The butterfly payoff has two wings around K

Long the calls at $K - h$ and $K + h$; short two calls at K.



$$f(S_T) = (S_T - K + h)^+ - 2(S_T - K)^+ + (S_T - K - h)^+ \geq 0.$$

NO-ARBITRAGE The outer strikes form the wings; the payoff peaks at the middle strike. Hence $C_t(K - h) - 2C_t(K) + C_t(K + h) \geq 0$.

The Law of One Price

Identical payoffs must have identical prices

Two self-financing strategies satisfy $X_T = Y_T$ in every state.

YOUR TURN If $X_0 \neq Y_0$, which strategy do you buy, and what are the cashflows?

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YOUR TURN If $X_0 \neq Y_0$, which strategy do you buy, and what are the cashflows?

Answer

Buy the cheaper strategy and sell the more expensive one. Receive the price gap today and owe $X_T - Y_T = 0$ at maturity. Therefore $\boxed{X_0 = Y_0}$.

Replication turns the law of one price into a pricing rule

Consequence for derivatives

If a payoff H is replicated by a self-financing strategy with initial cost X_0, then no-arbitrage forces

$$\boxed{V_0(H) = X_0.}$$

This is the bridge

Everything we called "martingale pricing" in Lecture 7 is this statement, combined with a convenient way of computing X_0 as a conditional expectation.

Can you rebuild put–call parity from the law of one price alone?

Compare a call plus K zero-coupon bonds with a put plus one share.

YOUR TURN Compute both terminal payoffs. What must their current costs satisfy?

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Answer

$$(S_T - K)^+ + K = (K - S_T)^+ + S_T = \max(S_T, K).$$

The payoffs coincide, so $\boxed{C_t + Ke^{-r(T-t)} = P_t + S_t}$.

What does a call, a put, and two strikes always cost, together?

For $K_1 < K_2$, buy $C(K_1)$ and $P(K_2)$, and sell $C(K_2)$ and $P(K_1)$.

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Answer

It pays $K_2 - K_1$ in every state, like a zero-coupon bond. Hence

$$\boxed{C_t(K_1) - C_t(K_2) + P_t(K_2) - P_t(K_1) = (K_2 - K_1)e^{-r(T-t)}.}$$

Application: Pricing a Forward Contract

A forward payoff can be replicated with a bond and a share

A forward pays $S_T - K_t$. A zero-coupon bond costs $P_t(T)$ and pays 1 at T. Assume no dividends.

YOUR TURN Replicate the forward payoff. Which K_t makes its initial value zero?

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YOUR TURN Replicate the forward payoff. Which K_t makes its initial value zero?

Answer

Buy one share and sell K_t bonds. Their cost is $S_t - K_t P_t(T)$, so

$$\boxed{K_t = \frac{S_t}{P_t(T)}.}$$

What changes if the share pays a continuous dividend yield q?

YOUR TURN Reinvested dividends turn $e^{-q(T-t)}$ shares today into one share at T. What is the zero-value forward delivery price?

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Answer

The prepaid share costs $S_t e^{-q(T-t)}$. Thus

$$\boxed{F_{t,T} = \frac{S_t e^{-q(T-t)}}{P_t(T)}}; \qquad P_t(T) = e^{-r(T-t)} \implies \boxed{F_{t,T} = S_t e^{(r-q)(T-t)}}.$$

From No-Arbitrage to a Probability Measure

No-arbitrage has a probabilistic face

Informal principle

A market is arbitrage-free precisely when there exists a probability measure under which discounted traded asset prices are martingales. Such a measure is called an equivalent martingale measure (EMM).

Already familiar

This is exactly the role $\mathbb{Q}$ played in Black–Scholes: under $\mathbb{Q}$, $D_t S_t$ is a martingale, and Girsanov is the tool that manufactures $\mathbb{Q}$ from $\mathbb{P}$.

Equivalence means $\mathbb{Q}$ and $\mathbb{P}$ agree on what is possible

$$\mathbb{Q} \sim \mathbb{P} \quad \Longleftrightarrow \quad \mathbb{Q} \text{ and } \mathbb{P} \text{ have the same null events.}$$

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Answer

If $\mathbb{P}(\omega) > 0$ but $\mathbb{Q}(\omega) = 0$, $\mathbb{Q}$ assigns zero value to a payoff occurring only in that genuinely possible state. Equivalence prevents pricing from ignoring such a risk.

The First Fundamental Theorem of Asset Pricing

FTAP I: no-arbitrage and existence of an EMM are the same statement

First Fundamental Theorem, informal

A financial market is arbitrage-free if and only if there exists an equivalent martingale measure $\mathbb{Q}$, under which every discounted traded asset price is a $\mathbb{Q}$-(local) martingale.

no arbitrage $\iff$ an EMM exists.

One theorem, two languages

Economic statement	Probabilistic statement
No free lunch from trading	An equivalent martingale measure exists
Prices are internally consistent	Discounted traded gains have no drift under $\mathbb{Q}$

Why it matters

The theorem lets us *replace* a hard combinatorial search for arbitrage strategies with an algebraic search for a consistent set of probabilities.

Is this one-period market arbitrage-free?

A stock has $S_0 = 100$, factors $u = 1.2$, $d = 0.8$, and the bond has gross return $R = 1.1$.

YOUR TURN Find the up probability p that makes the discounted stock a $\mathbb{Q}$-martingale. Is it a valid probability?

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Answer

$$pu + (1-p)d = R \quad \Longrightarrow \quad \boxed{p = \frac{R-d}{u-d} = 0.75 \in (0,1).}$$

An equivalent martingale measure exists. In general this requires $d < R < u$.

What if the risk-free rate is too good, $R = 1.25 > u$?

Keep $S_0 = 100$, $u = 1.2$, and $d = 0.8$, but set $R = 1.25$.

YOUR TURN Since no $p \in (0, 1)$ makes the discounted stock a martingale, construct a zero-cost trade that profits in both states.

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Answer

Short one share for 100 and invest the proceeds in the bond. At T, receive 125, buy back the share, and keep

$$\boxed{125 - S_T \in \{5, 45\} > 0.}$$

A state price buys one dollar in one state

The state claim $e_k(\omega_j) = \mathbf{1}_{\{j=k\}}$ pays 1 only if ω_k occurs. Its positive state price ψ_k is today's cost assigned to that payoff.

$$\boxed{V_0(H) = \sum_{k=1}^{K} \psi_k H(\omega_k)} \qquad \text{for every traded or attainable payoff } H.$$

If a bond pays 1 at T and costs $B_0 = 1/R$, then

$$\sum_k \psi_k = B_0, \qquad q_k = R\psi_k > 0, \qquad \sum_k q_k = 1.$$

Thus $V_0(H) = R^{-1}\mathbb{E}^{\mathbb{Q}}[H]$, where $\mathbb{Q}(\omega_k) = q_k$.

When a state claim is not traded

The ψ_k are consistent shadow prices; they need not be unique in an incomplete market.

A bond and stock determine two state prices

The bond pays 1 in each state and costs $B_0 = 1/1.1$; the stock costs 100 and pays 120 (up) or 90 (down).

$$\psi_1 + \psi_2 = \frac{1}{1.1}, \qquad 120\psi_1 + 90\psi_2 = 100 \quad \Longrightarrow \quad \boxed{\psi_1 \approx 0.6061, \quad \psi_2 \approx 0.3030.}$$

Both are positive. Multiplying by the bond return gives the risk-neutral probabilities:

$$q_{\text{up}} = 1.1\psi_1 = \tfrac{2}{3}, \qquad q_{\text{down}} = 1.1\psi_2 = \tfrac{1}{3}.$$

Reading

Each state price is a discounted risk-neutral probability: $\psi_k = B_0 q_k$.

Can a quoted stock price be inconsistent with the bond and the payoffs?

A bond paying 1 costs $B_0 = 1/1.1$. A stock quoted at 115 pays 120 or 90 at maturity.

YOUR TURN Solve for the two state prices. If one is negative, exhibit the trade.

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A bond paying 1 costs $B_0 = 1/1.1$. A stock quoted at 115 pays 120 or 90 at maturity.

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Answer

$\psi_1 + \psi_2 = 1/1.1$, $120\psi_1 + 90\psi_2 = 115$ give $\psi_1 \approx 1.106$, $\psi_2 \approx -0.197$. Short the stock for 115, buy 120 bonds for $120/1.1$, and keep the initial surplus. At T, $120 - S_T \geq 0$.

The Second Fundamental Theorem of Asset Pricing

FTAP II: completeness and uniqueness of the EMM are the same statement

Second Fundamental Theorem, informal

In an arbitrage-free market, the market is complete (every claim is replicable) if and only if the equivalent martingale measure is unique.

$$\boxed{\text{market complete} \iff \text{EMM unique.}}$$

- Unique EMM $\implies$ every replicable claim has one unambiguous price.
- Several EMMs $\implies$ no-arbitrage alone leaves a range of admissible prices for unreplicable claims.

Does one bond and one stock pin down a three-state model?

Let $R = 1.1$, $S_0 = 100$, and $S_T \in \{130, 100, 70\}$ in states (u, m, d).

YOUR TURN Write the risk-neutral equations. Are the three probabilities unique?

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Answer

$$p_u + p_m + p_d = 1, \qquad 1.3p_u + p_m + 0.7p_d = 1.1.$$

With $p_m = t$, the solutions are $\boxed{p_u = \frac{2}{3} - \frac{1}{2}t, \quad p_d = \frac{1}{3} - \frac{1}{2}t}$ for $0 < t < 2/3$: the EMM is not unique.

Different EMMs price an unreplicable claim differently

Take $t = 0.2$ and $t = 0.4$, both valid:

$$t = 0.2: \ (p_u, p_m, p_d) = (0.5667,\ 0.2,\ 0.2333), \qquad t = 0.4: \ (0.4667,\ 0.4,\ 0.1333).$$

YOUR TURN Consider a claim paying 1 only in the middle state m. Compute its price under each measure. Do they agree?

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YOUR TURN Consider a claim paying 1 only in the middle state m. Compute its price under each measure. Do they agree?

$$\text{price} = \frac{p_m}{R}: \qquad \frac{0.2}{1.1} \approx 0.182, \qquad \frac{0.4}{1.1} \approx 0.364.$$

Two legitimate EMMs, two different prices: no-arbitrage alone cannot select one.

Can a third traded asset restore completeness?

In the three-state market, add a call paying $(30, 0, 0)$, quoted at $C_0 \approx 13.64$.

YOUR TURN Use its pricing equation to find p_u; then find p_m, p_d.

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Answer

$$\frac{30p_u}{1.1} = 13.64 \quad \Longrightarrow \quad p_u \approx \tfrac{1}{2}.$$

The earlier two equations give $\boxed{(p_u, p_m, p_d) \approx (\tfrac{1}{2}, \tfrac{1}{3}, \tfrac{1}{6})}$. The independent call payoff makes the EMM unique.

Completeness is about spanning, not about counting assets

Why it worked

The payoff vectors of the bond $(1, 1, 1)$, the stock $(1.3, 1.0, 0.7)$, and the call $(30, 0, 0)$ are linearly independent: three genuinely different exposures for three states.

NO-ARBITRAGE This is literally why traders add options to a hedging book: a derivative with a payoff the existing assets cannot replicate expands the span of attainable claims, exactly as FTAP II predicts.

Superhedging and Price Bounds

When exact replication fails, dominance still gives bounds

A self-financing strategy with wealth X *superhedges* H if $X_T \geq H$ a.s.

$$\boxed{V_0(H) \leq X_0.}$$

A subhedging strategy $Y_T \leq H$ gives a lower bound:

$$\boxed{Y_0 \leq V_0(H).}$$

Exact replication is the special case

When $X_T = H$ exactly, the upper and lower bounds coincide and $V_0(H) = X_0$.

Parity plus positivity gives the call its lower bound for free

We already derived $C_t - P_t = S_t - Ke^{-r(T-t)}$ from the law of one price.

YOUR TURN Combine this identity with $P_t \geq 0$ and $(S_T - K)^+ \leq S_T$ to bound C_t above and below, with no distributional assumption at all.

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$$\boxed{\left(S_t - Ke^{-r(T-t)}\right)^+ \leq C_t \leq S_t.}$$

Can you bound a best-of-two-stocks claim without a model?

Let $H = \max(S_T^1, S_T^2)$, with nonnegative stock prices.

YOUR TURN Which stock portfolio lies below H, and which lies above it?

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Let $H = \max(S_T^1, S_T^2)$, with nonnegative stock prices.

YOUR TURN Which stock portfolio lies below H, and which lies above it?

Answer

Either single stock pays at most H; holding both pays at least H. Dominance gives

$$\boxed{\max(S_t^1, S_t^2) \leq V_t(H) \leq S_t^1 + S_t^2.}$$

Can a digital option be bounded using only call prices?

A digital pays $D = \mathbf{1}_{\{S_T > K\}}$. For $h > 0$, compare it with the bull spread at $K, K + h$.

YOUR TURN Does $\left[(S_T - K)^+ - (S_T - K - h)^+\right]/h$ lie above or below D?

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Answer

The spread pays 0, then $(S_T - K)/h$, then 1; it never exceeds D. Therefore

$$\boxed{\frac{C_t(K) - C_t(K+h)}{h} \leq V_t(D).}$$

Can you build the matching superhedge?

YOUR TURN Compare the digital $D = \mathbf{1}_{\{S_T > K\}}$ with the bull spread at $K - h, K$, scaled by $1/h$. Which bound follows?

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Answer

This spread dominates D, so its price is an upper bound. Together with the previous slide,

$$\boxed{\frac{C_t(K) - C_t(K+h)}{h} \leq V_t(D) \leq \frac{C_t(K-h) - C_t(K)}{h}.}$$

Synthesis

Four ideas, stacked in order

1. **No-arbitrage principles** (dominance, law of one price) rule out inconsistent prices, with no model.
2. **Replication** gives an exact price whenever a payoff can be synthesized from traded assets.
3. **Equivalent martingale measures** translate no-arbitrage into a probabilistic statement amenable to computation.
4. **Completeness** determines whether that computation returns one price or only a range.

The big picture in one line

no-arbitrage $\iff$ martingale pricing under a suitable measure.

This lecture versus Lecture 7

Lecture 7 *used* an EMM to price and hedge. Today explains its existence (FTAP I), its uniqueness (FTAP II), and the bounds that remain when replication is unavailable.

Can you diagnose an arbitrage from raw quotes?

For one maturity, suppose $C_t(90) = 14$, $C_t(100) = 12$, and $C_t(110) = 7$.

YOUR TURN Check strike monotonicity and convexity. Construct the trade if a quote violates either condition.

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Answer

Monotonicity holds, but convexity requires $C_t(100) \leq (14 + 7)/2 = 10.5 < 12$. Buy the wings and sell two middle calls: cost $14 + 7 - 2(12) = -3$, while the butterfly payoff is nonnegative in every state.

Five checks before trusting a claimed no-arbitrage price

1. Is the comparison payoff actually dominating, in *every* state?
2. Is the replicating strategy genuinely self-financing?
3. Is the candidate pricing measure equivalent to $\mathbb{P}$, not merely a probability measure?
4. Is the EMM unique for this market, or only for the traded assets?
5. If replication fails, have you produced *both* a superhedge and a subhedge, not just one?

NO-ARBITRAGE Dominance, the law of one price, and the two Fundamental Theorems are the same idea seen at four levels of abstraction.